CUDEVICES

SERIES: TS04 | DESCRIPTION: TACTILE SWITCH

FEATURES

- surface mount
- black actuator



RoHS

SPECIFICATIONS

parameter	conditions/description	min	typ	max	units
rated voltage		1		12	Vdc
rated current		0.01		50	mA
withstanding voltage	for 1 minute		250		Vac
contact resistance	applying load of 2 times operating force at 5 Vdc, 10 mA			100	mΩ
insulation resistance	at 100 Vdc for 1 minute	100			MΩ
operating force	see force table				
actuator travel		0.15	0.25	0.35	mm
operating temperature		-30		80	°C
storage temperature		-40		90	°C
life	rated load, 2 cycles/second, 1.5 times operating force 100 gf models 160 gf models 260 gf models		 100,000 80,000 60,000		 cycles cycles cycles
vibration	10~55~10 Hz, 1.5 mm amplitude, 2 hours on each XYZ				
flammability rating	see material table				
RoHS	yes				

PART NUMBER KEY

TS04 - 66 - XX - BK - XXX - SMT - XX

Base Number

Actuator Height:
43 = 4.3 mm
50 = 5.0 mm
55 = 5.5 mm
65 = 6.5 mm
70 = 7.0 mm
*73 = 7.3 mm
75 = 7.5 mm
85 = 8.5 mm
95 = 9.5 mm

Push Force:
100 = 100 gf
160 = 160 gf
260 = 260 gf

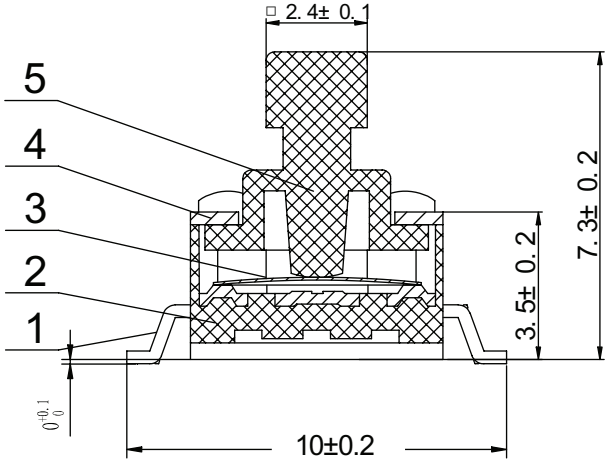
Packaging:
"blank" = bag
TR = tape and reel

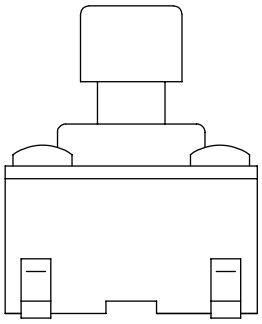
Note: *7.3 mm height versions have square actuator top. All other models have round actuator top.

MECHANICAL DRAWING (7.3 MM HEIGHT MODELS)

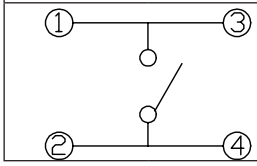
units: mm
tolerance: ±0.2 mm
unless otherwise noted

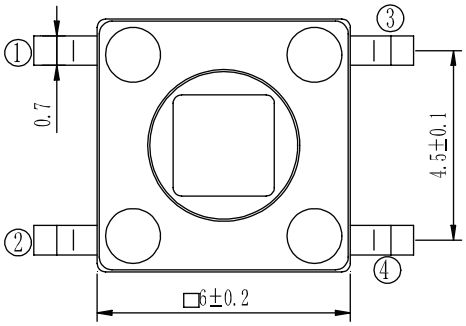
ITEM	DESCRIPTION	MATERIAL	PLATING/COLOR
1	terminals	brass	silver
2	case	PA66 (UL94HB)	black
3	contacts	stainless steel	silver
4	cover	stainless steel	
5	stem	PA66 (UL94HB)	black

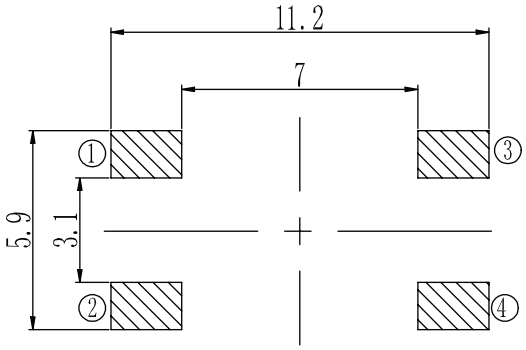




ACTUATOR POSITION [ ]: MOMENTARY	
NORMAL	DOWN
OFF	[ON]
SCHEMATIC	







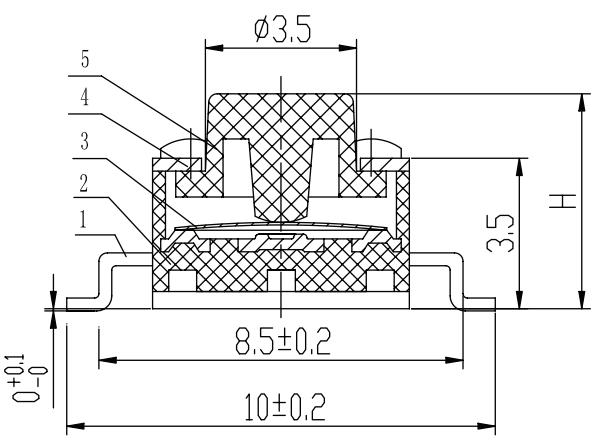
Recommended PCB Layout
Top View

Operting Force Table		
Model	Press Force (N)	Return Force (N)
100	1.0±0.3	>0.2
160	1.6±0.5	>0.2
260	2.6±0.5	>0.2

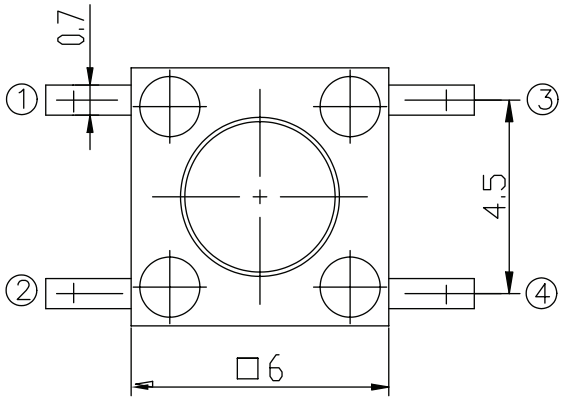
MECHANICAL DRAWING (ALL OTHER HEIGHT MODELS)

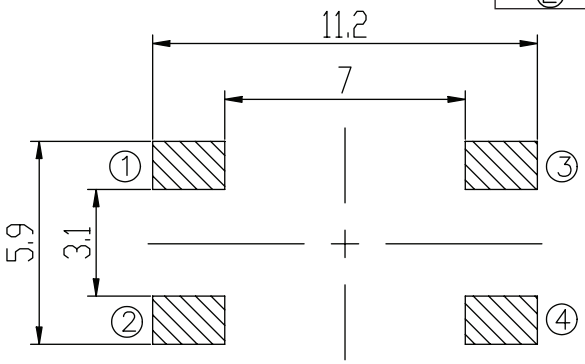
units: mm
tolerance: ±0.2 mm
unless otherwise noted

ITEM	DESCRIPTION	MATERIAL	PLATING/COLOR
1	terminals	brass	silver
2	case	PA66 (UL94HB)	black
3	contacts	stainless steel	silver
4	cover	stainless steel	
5	stem	PA66 (UL94HB)	black



ACTUATOR POSITION [ ]: MOMENTARY	
NORMAL	DOWN
OFF	[ON]
SCHEMATIC	





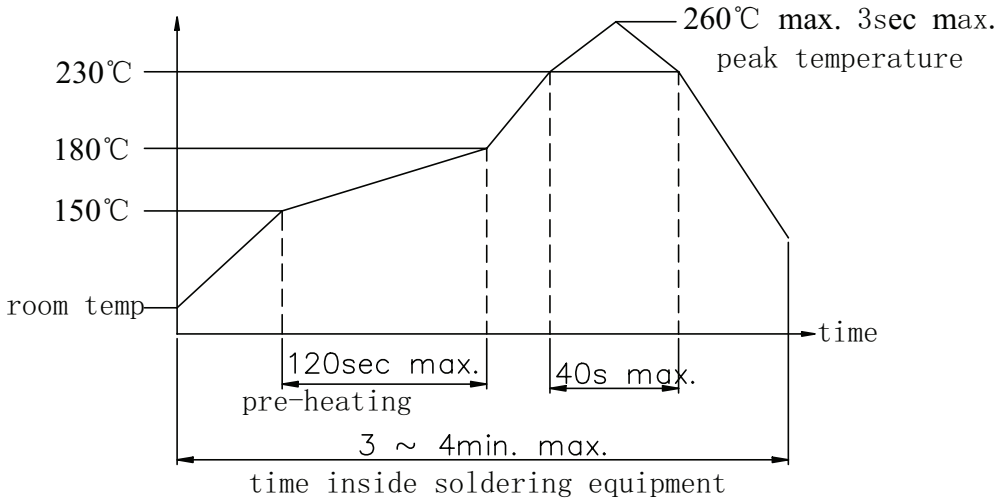
Recommended PCB Layout
Top View

Operting Force Table		
Model	Press Force (N)	Return Force (N)
100	1.0±0.3	>0.2
160	1.6±0.5	>0.2
260	2.6±0.5	>0.2

SOLDERABILITY

parameter	conditions/description	min	typ	max	units
hand soldering	for maximum 3 seconds			350	°C
reflow soldering[1]	see reflow profile			260	°C

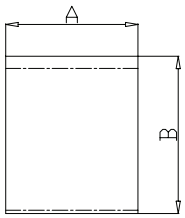
Notes: 1. Do not exceed 2 reflow cycles.



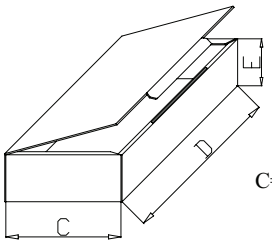
PACKAGING (BAG)

units: mm

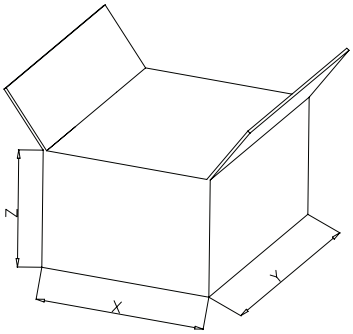
Bag Size: 190 x 160 mm
Carton Box Size: 445 x 350 x 280 mm
Bag QTY: 1,000 pcs per bag
Carton Box QTY: 60,000 pcs per carton



A=160mm、B=190mm



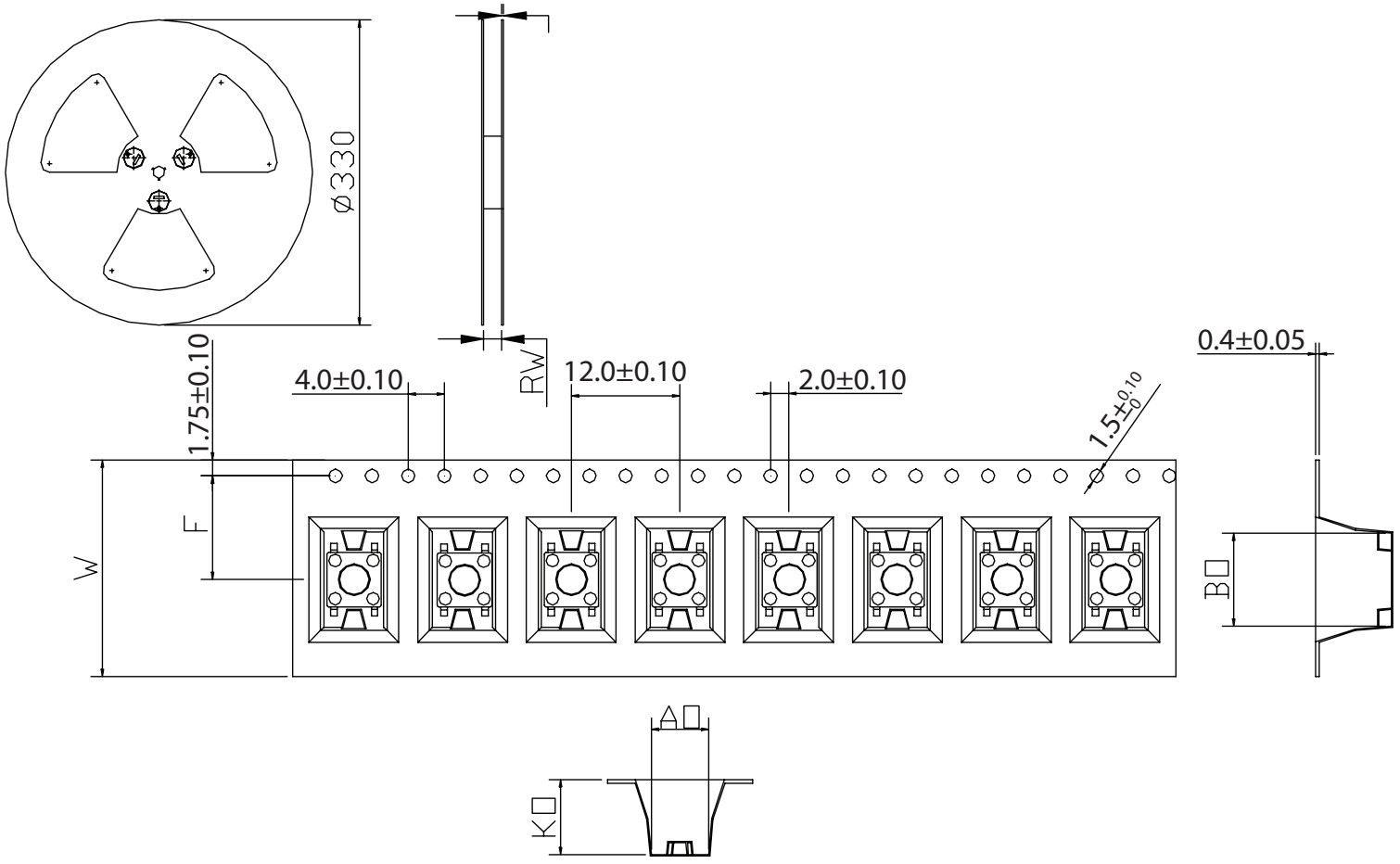
C=160mm、D=216mm、E=48mm



X=350mm、Y=455、Z=280mm

PACKAGING (TAPE & REEL)

Packaging Table									
Actuator Height Models (mm)	Reel QTY	Carton Size (XxYxZ) (mm)	Carton QTY	RW (mm)	W (mm)	A0 (mm)	B0 (mm)	K0 (mm)	F (mm)
4.3	1,000	340 x 430 x 340	20,000	16.5	16±0.3	6.3±0.1	10.4±0.1	4.5±0.1	7.5±0.1
5.0	1,000	340 x 430 x 340	20,000	16.5	16±0.3	6.3±0.1	10.4±0.1	5.2±0.1	7.5±0.1
5.5	1,000	340 x 430 x 340	20,000	16.5	16±0.3	6.3±0.1	10.4±0.1	5.7±0.1	7.5±0.1
6.5	800	340 x 430 x 340	16,000	16.5	16±0.3	6.3±0.1	10.4±0.1	6.7±0.1	7.5±0.1
7.0	750	340 x 430 x 340	10,500	24.5	24±0.3	6.3±0.1	10.3±0.1	7.3±0.1	11.5±0.1
7.3	750	340 x 430 x 340	10,500	24.5	24±0.3	6.4±0.1	10.5±0.1	7.8±0.1	11.5±0.1
7.5	750	340 x 430 x 340	10,500	24.5	24±0.3	6.4±0.1	10.5±0.1	7.8±0.1	11.5±0.1
8.5	500	340 x 430 x 340	7,000	24.5	24±0.3	6.3±0.1	10.4±0.1	$8.6^{+0.1}_{-0}$	11.5±0.1
9.5	500	340 x 430 x 340	7,000	24.5	24±0.3	6.3±0.1	10.4±0.1	$9.6^{+0.1}_{-0}$	11.5±0.1



Note: 1. Actuator height 4.3 & 5.0 models have about 15~25 open positions reserved for the front and rear positions of the reel.
2. Actuator height 5.5 & 6.5 models have about 15~30 open positions reserved for the front and rear positions of the reel.
3. Actuator height 7.0, 7.3, & 7.5 models have about 20~35 open positions reserved for the front and rear positions of the reel.
4. Actuator height 8.5 & 9.5 models have about 30~50 open positions reserved for the front and rear positions of the reel.

REVISION HISTORY

rev.	description	date
1.0	initial release	09/20/2021
1.01	added 7.3 mm height model drawing	11/03/2021
1.02	logo, datasheet style update	08/05/2022
1.03	added tape and reel packaging option	05/12/2023

The revision history provided is for informational purposes only and is believed to be accurate.

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